

KIRKUK, Iraq

WMO: 406210

Lat: 35.469N

Lon: 44.349E

Elev: 1061

StdP: 14.14

Time Zone: 3.00 (E03)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	34.8	37.5	12.4	10.8	52.0	16.2	13.0	56.6	22.7	54.7	18.6	55.2	3.7	60	0.433

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.6	114.5	70.0	112.0	69.4	109.8	69.0	75.2	107.0	73.4	105.3	71.9	103.4	8.2	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.2	93.4	91.9	61.8	85.9	87.2	60.1	80.5	86.3	39.4	106.2	37.8	105.3	36.3	103.6	84.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.8	14.8	12.7	DB	29.9	117.7	2.4	2.2	28.2	119.3	26.8	120.6	25.5	121.8	23.8	123.4	(4)
(5)			WB	28.3	76.1	3.8	5.1	25.6	79.8	23.3	82.8	21.2	85.7	18.4	89.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	74.2	49.3	53.1	60.3	70.2	82.8	93.3	98.2	97.9	89.5	77.7	62.4	54.5	(6)	
	DBStd	18.28	5.51	5.94	6.28	6.92	6.54	4.59	3.83	3.52	5.36	6.81	5.82	5.71	(7)	
	HDD50	138	79	33	3	0	0	0	0	0	0	0	1	22	(8)	
	HDD65	1466	486	333	175	24	0	0	0	0	0	2	116	330	(9)	
	CDD50	8979	58	121	322	605	1018	1300	1494	1484	1186	859	372	160	(10)	
	CDD65	4836	0	1	29	180	553	850	1029	1019	736	397	38	4	(11)	
	CDH74	76591	0	3	230	1788	7661	14412	18325	18082	11384	4416	261	29	(12)	
(13)	CDH80	52492	0	0	43	684	4410	10208	13868	13621	7464	2147	44	3	(13)	
(14)	Wind	WSAvg	5.2	4.6	4.9	5.3	5.6	6.0	6.0	5.7	5.0	4.8	4.2	4.3	(14)	
(15) Precipitation	PrecAvg	18.7	2.3	2.4	2.6	2.7	1.6	0.5	0.5	0.3	0.4	1.3	2.0	2.3	(15)	
	PrecMax	25.0	3.7	4.5	5.2	4.8	4.7	1.5	1.0	0.7	1.3	3.2	4.7	4.7	(16)	
	PrecMin	11.2	0.9	0.6	1.1	0.9	0.4	0.1	0.1	0.1	0.0	0.2	0.2	0.5	(17)	
	PrecStd	3.3	0.8	0.9	1.1	1.1	0.9	0.4	0.2	0.2	0.3	0.8	1.2	1.2	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	67.8	73.8	84.5	98.4	106.0	113.2	117.8	116.9	111.4	100.4	84.4	77.3	(19)	
		MCWB	52.9	56.1	56.9	62.6	65.8	71.2	70.3	70.0	68.8	67.5	61.5	53.2	(20)	
	2%	DB	64.1	69.8	79.1	91.8	102.4	110.8	114.8	114.5	108.2	97.0	80.5	71.2	(21)	
		MCWB	51.2	54.3	57.3	63.0	65.8	68.9	70.2	69.7	68.3	64.5	57.9	53.2	(22)	
	5%	DB	61.1	66.4	75.6	87.5	99.0	108.0	112.7	111.6	105.2	93.8	76.8	66.4	(23)	
		MCWB	50.2	52.6	55.2	61.1	64.9	67.3	69.6	69.4	67.7	63.5	57.0	52.1	(24)	
	10%	DB	58.9	63.6	72.0	83.9	96.5	105.9	110.4	109.6	102.1	91.0	73.4	63.6	(25)	
		MCWB	49.8	51.2	54.7	60.5	64.3	66.7	69.0	69.5	67.2	62.6	55.7	51.5	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	56.8	59.0	62.5	68.4	72.2	78.0	78.6	76.9	74.9	71.5	67.1	58.5	(27)	
		MCDB	61.7	67.4	74.8	85.9	96.0	107.1	108.5	109.5	104.6	93.1	81.2	65.7	(28)	
	2%	WB	54.5	57.0	60.3	66.0	69.6	73.5	74.7	74.8	71.8	68.8	63.4	56.9	(29)	
		MCDB	60.7	65.8	72.4	82.8	96.2	103.5	107.2	107.4	101.7	91.6	73.9	63.2	(30)	
	5%	WB	52.6	54.7	58.3	64.3	67.7	71.0	72.8	73.4	70.2	66.8	61.1	55.5	(31)	
		MCDB	59.0	63.5	70.7	80.4	91.5	101.2	106.2	106.2	99.8	86.2	69.2	62.3	(32)	
	10%	WB	50.9	52.8	56.6	62.6	66.1	68.9	71.2	72.0	68.8	65.0	58.5	53.9	(33)	
		MCDB	56.9	61.1	67.6	78.2	89.7	100.2	105.1	103.8	98.2	83.2	67.4	60.9	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	15.2	16.0	17.7	19.9	21.2	22.9	23.6	23.6	23.2	20.4	18.5	15.9	(35)	
		MCDBR	17.5	18.9	21.9	24.0	23.8	24.7	25.0	25.0	25.3	23.0	22.4	20.3	(36)	
	5% WB	MCWBR	9.8	9.6	9.7	9.3	8.2	8.7	8.5	8.6	9.1	9.0	10.5	10.6	(37)	
		MCWBR	14.0	15.9	17.6	20.4	21.4	22.2	23.5	23.5	23.4	19.3	16.2	14.6	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.405	0.436	0.478	0.563	0.562	0.501	0.528	0.489	0.452	0.473	0.409	0.400	(40)	
		taud	2.048	1.973	1.888	1.725	1.742	1.869	1.815	1.904	1.981	1.935	2.079	2.072	(41)	
	Ebn at Noon		240	248	249	235	238	252	244	251	253	232	237	232	(42)	
		Edh at Noon	41	50	59	73	73	64	67	60	53	51	40	38	(43)	
(44) All-Sky Solar Radiation	RadAvg		830	1102	1504	1808	2186	2539	2425	2218	1862	1336	990	760	(44)	
	RadStd		55	99	120	102	115	100	63	60	81	98	97	86	(45)	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page